LIBRARY MANAGEMENT SYSTEM

UML Class Diagram Report

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1. Introduction

This report presents the UML class model for the proposed Library Management System. It identifies the system's main classes, attributes, methods, and relationships.

2. UML Class Diagram

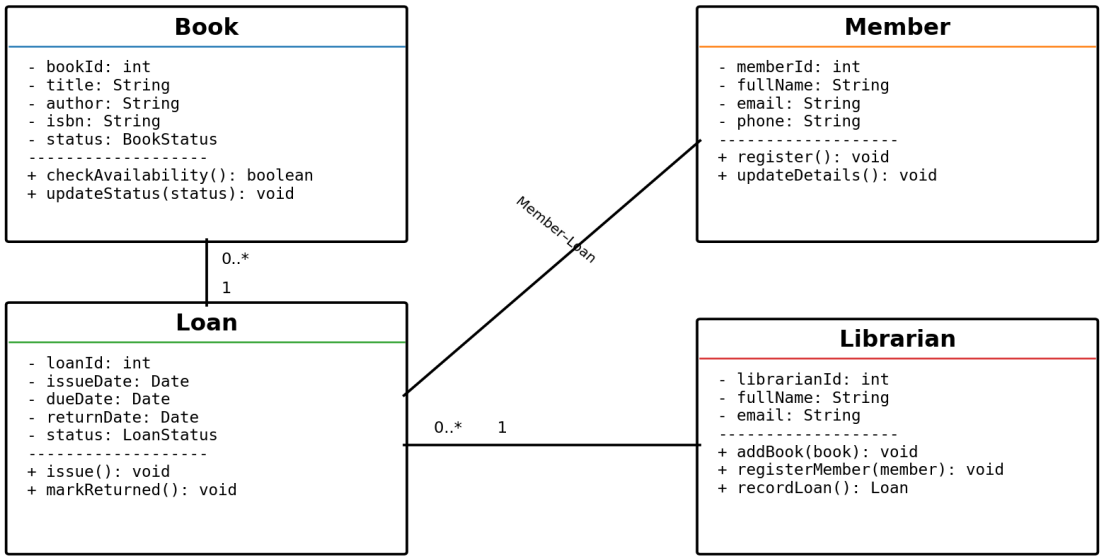


Figure 1. Proposed UML class diagram for the Library Management System.

3. Classes, Attributes, and Methods

Book

Attributes: bookId, title, author, isbn, status.
Methods: checkAvailability(), updateStatus(status).

Member

Attributes: memberId, fullName, email, phone.
Methods: register(), updateDetails().

Loan

Attributes: loanId, issueDate, dueDate, returnDate, status.
Methods: issue(), markReturned().

Librarian

Attributes: librarianId, fullName, email.
Methods: addBook(book), registerMember(member), recordLoan().

4. Relationships and Multiplicities

• Member–Loan (1 to 0..*): one member can have zero or many loan records; each loan belongs to one member.
• Book–Loan (1 to 0..*): a book can appear in many loan records over time; each loan refers to one book.
• Librarian–Loan (1 to 0..*): one librarian can record zero or many loans; each loan is recorded by one librarian.

5. Design Assumptions

Each Loan record refers to exactly one Book, one Member, and one Librarian. A book may have multiple historical loan records, while the availability status indicates whether it can currently be borrowed. The model does not define database tables or user-interface screens.

6. Conclusion

The proposed class model represents the core structure of a Library Management System. It separates book data, member data, borrowing transactions, and librarian responsibilities. The model can be refined later by adding fine management, reservations, or multiple physical copies of the same title.